

STT 465
Bayesian Data Analysis
Lecture 03: Probability Review II

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Learning Goals

- A working understanding of probability theory
 - Probability
 - Bayes' rule
 - Generating random samples in R
 - Random variable (r.v.)
 - Common distributions
 - Generating r.v.'s in R

From Probability to Random Variable

- Consider an experiment in which two coins are flipped together
 - The sample space $\Omega = \{TT, TH, HT, HH\}$
 - The event space $\mathcal{F} = \{\emptyset, \{TT\}, \dots, \{HH\}, \{TT, TH\}, \dots, \Omega\}$
 - The probability measure P
- What is the probability of the number of heads equals to x ?
 - $P(X = 0) = P(A_{TT})$
 - $P(X = 1) = P(A_{TH}) + P(A_{HT})$
 - $P(X = 2) = P(A_{HH})$

Defining Random Variable

- A random variable (r.v.) X is a function

$$X : \Omega \rightarrow \mathbb{R}$$

- Denote r.v. using a uppercase letter $X(\omega)$ or more simply X
- Denote the value that a r.v. may take on using a lowercase letter x

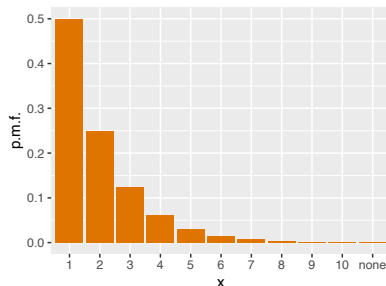
Examples

- Consider an experiment in a coin is flipped 10 times, where $P(H) = q$ and $P(T) = 1 - q$
 - Suppose the outcome is $\omega_0 = \{TTHHTHTTHHH\}$
- A r.v. is $X : \Omega \rightarrow \mathbb{R}$
 - Number of tosses until a head occurs: $X(\omega_0) = 3$
 - Number of heads: $X(\omega_0) = 5$
- Measure r.v. using probability
 - What is the probability of observing the first head in the x -th flip?
 - What is the probability of observing x heads in all 10 flips?

Examples

- Suppose the coin is fair, i.e. $P(H) = 0.5$ and $P(T) = 0.5$
- What is the probability of observing the first head in the x -th toss?
- By a table
- By a bar plot

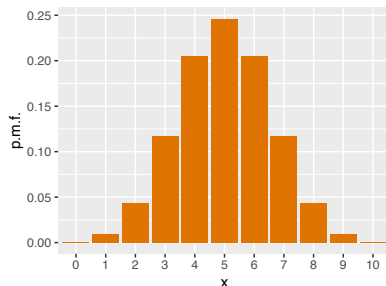
x	Probability
1	0.500
2	0.250
3	0.125
4	0.062
5	0.031
6	0.016
7	0.008
8	0.004
9	0.002
10	0.001
> 10	0.001



Examples

- Suppose the coin is fair, i.e. $P(H) = 0.5$ and $P(T) = 0.5$
- What is the probability of observing x heads in all 10 tosses?
- By a table
- By a bar plot

x	Probability
0	0.001
1	0.010
2	0.044
3	0.117
4	0.205
5	0.246
6	0.205
7	0.117
8	0.044
9	0.010
10	0.001



Probability Mass Function

- If a r.v. X takes on a finite set of possible values
 - X is a discrete r.v.
 - The probability mass function (p.m.f.) is a function $p_X : \mathbb{N} \rightarrow [0, 1]$ such that

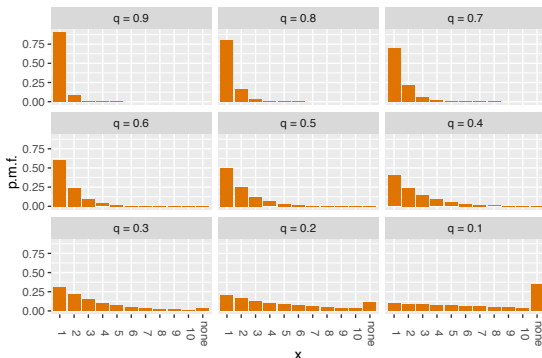
$$p_X(x) = P(X = x)$$

- $p_X(x)$ should satisfy:
 - $0 \leq p_X(x) \leq 1$
 - $\sum_{x \in \text{Val}(X)} p_X(x) = 1$, where $\text{Val}(X)$ is the set of all possible values that X can take
 - $\sum_{x \in A} p_X(x) = P(X \in A)$

Probability Mass Function

- Geometric distribution

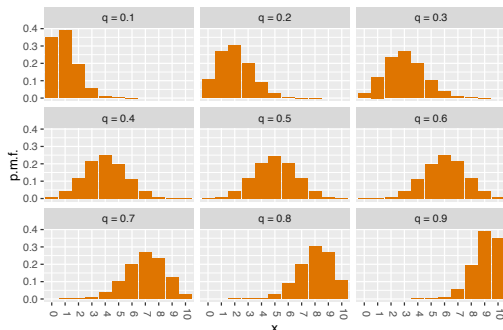
- Gives the probability that the first occurrence of success requires x independent trials (i.e. **Bernoulli trials**), each with success probability q
- p.m.f.: $p_X(x; q) = (1 - q)^{x-1}q$, where $x = 1, 2, \dots$



Probability Mass Function

- Binomial distribution

- Gives the probability that the number of successes in a sequence of n independent trials (i.e. **Bernoulli trials**), each with success probability q
- p.m.f.: $p_X(x; q, N) = \binom{N}{x} q^x (1 - q)^{N-x}$, where $x = 0, 1, \dots, N$



Cumulative Distribution Function

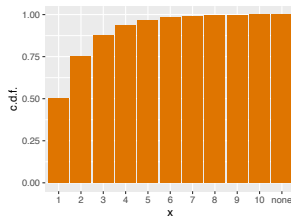
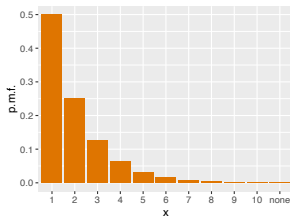
- To specify the probability measures used when dealing with r.v., it is often convenient to specify an alternative function
- A cumulative distribution function (c.d.f.) is a function $F_X : \mathbb{R} \rightarrow [0, 1]$ such that

$$F_X(x) = P(X \leq x)$$

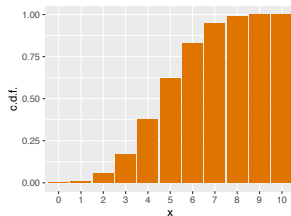
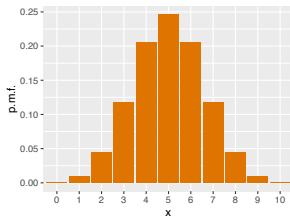
- $P(a < X \leq b) = F_X(b) - F_X(a)$, where a and b are two real values and $a \leq b$
- $F_X(x)$ should satisfy:
 - $0 \leq F_X(x) \leq 1$
 - $\lim_{x \rightarrow -\infty} F_X(x) = 0$
 - $\lim_{x \rightarrow \infty} F_X(x) = 1$
 - $x \leq y \Rightarrow F_X(x) \leq F_X(y)$

Cumulative Distribution Function

- Geometric distribution, where $q = 0.5$

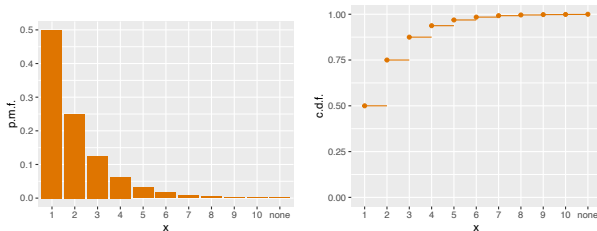


- Binomial distribution, where $q = 0.5$ and $n = 10$

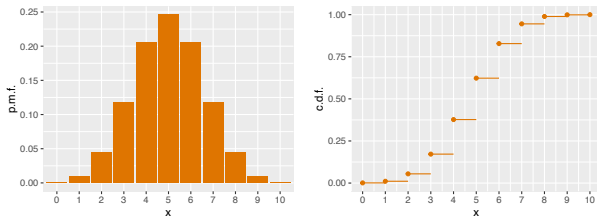


Cumulative Distribution Function

- Geometric distribution, where $q = 0.5$



- Binomial distribution, where $q = 0.5$ and $n = 10$



Probability Density Function

- If a r.v. X takes on an infinite number of possible values
 - X is a continuous r.v.
 - The probability density function (p.d.f.) is a function $f_X : \mathbb{R} \rightarrow \mathbb{R}_+$
- For some continuous r.v., the c.d.f. $F_X(x)$ is differentiable everywhere, the p.d.f. is defined as the derivative of the c.d.f.:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Probability Density Function

- According to the differentiation properties, for very small Δx ,

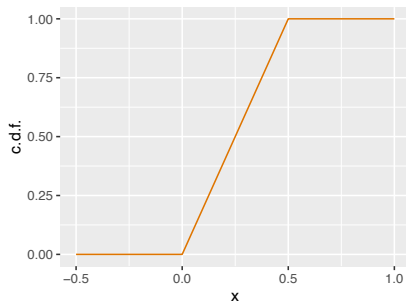
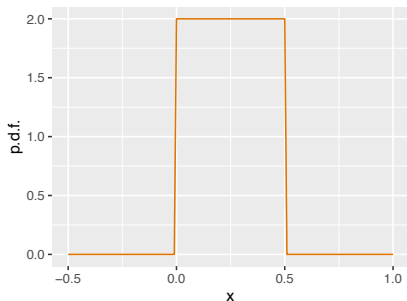
$$P(x \leq X \leq x + \Delta x) \approx f_X(x)\Delta x$$

- The value of p.d.f. at any given point x is not the probability of that event, i.e. $f_X(x) \neq P(X = x)$
 - $P(X = x)$ can only take any value between 0 and 1
 - $f_X(x)$ can take any value larger than 0
- $f_X(x)$ should satisfy:
 - $f_X(x) \geq 0$
 - $\int_{-\infty}^{\infty} f_X(x) = 1$
 - $\int_{x \in A} f_X(x)dx = P(X \in A)$

Probability Density Function

- Uniform distribution (Unif)

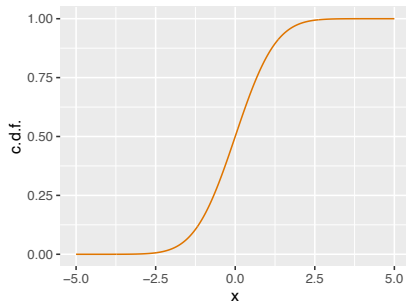
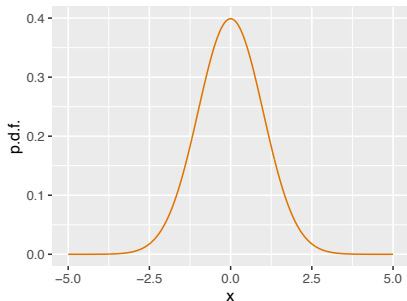
- All intervals of the same length on the distribution's support are equally probable
- p.d.f.: $f_X(x; a, b) = \frac{1}{b-a}$, where $a \leq x \leq b$



- Example: Waiting time for evenly operated bus service

Probability Density Function

- Normal distribution (also known as Gaussian distribution)
 - The most common continuous probability distribution
 - p.d.f.: $f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, where $x \in \mathbb{R}$



- Example: Difference between actual and scheduled arrival time for a bus

Discrete vs. Continuous r.v.

	Discrete	Continuous
# of possible Val	Countable	Uncountable
Definition on Ω	$P(X = x) =$ $P(\{\omega X(\omega) = x\})$	$P(a \leq X \leq b) =$ $P(\{\omega a \leq X(\omega) \leq b\})$
Form	p.m.f.: $P_X : \mathbb{N} \rightarrow [0, 1]$	p.d.f.: $f_X : \mathbb{R} \rightarrow \mathbb{R}_+$
Properties	$0 \leq p_X(x) \leq 1$ $\sum_{x \in \text{Val}(X)} p_X(x) = 1$ $\sum_{x \in A} p_X(x) = P(X \in A)$	$f_X(x) \geq 0$ $\int_{-\infty}^{\infty} f_X(x) dx = 1$ $\int_{x \in A} f_X(x) dx = P(X \in A)$

Expectation

- The expected value of a r.v., written as $E[X]$, is the most key aspect of its probability distribution
- Represents the average of a large number of independent realizations of the r.v.
- Also known as mean or first moment

Expectation

- Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary function
- Suppose X is a discrete r.v. with p.m.f. $p_X(x)$, then the expectation of $g(X)$ is defined as

$$\mathbb{E}[g(X)] = \sum_{x \in \text{Val}(X)} g(x)p_X(x)$$

- Suppose X is a continuous r.v. with p.d.f. $f_X(x)$, then the expectation of $g(X)$ is defined as

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- If $g(X) = X$, then the expectation is the mean of the r.v. X

Expectation

- Properties of expectation:

- $E[a] = a$ for any constant $a \in \mathbb{R}$
- $E[ag(X)] = aE[g(X)]$ for any constant $a \in \mathbb{R}$
- Linearity of expectation: $E[ag(X) + bf(Y)] = aE[g(X)] + bE[f(Y)]$ for any constant $a, b \in \mathbb{R}$

Variance

- The variance of a r.v, written as $\text{Var}(X)$, is the second key aspect of a probability distribution
- Measures how concentrated the distribution of a r.v. X is around its mean, i.e. $E[X]$
- Also known as second central moment $\text{Var}(X) = E[(X - E[X])^2]$
- An alternative expression

$$\begin{aligned} E[(X - E[X])^2] &= E[X^2 - 2E[X]X + E[X]^2] \\ &= E[X^2] - 2E[X]E[X] + E[X]^2 \\ &= E[X^2] - E[X]^2 \end{aligned}$$

Variance

- Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary function
- Suppose X is a discrete r.v. with p.m.f. $p_X(x)$, then the variance of $g(X)$ is defined as

$$\text{Var}(g(X)) = \sum_{x \in \text{Val}(X)} (g(x) - \mathbb{E}[g(X)])^2 p_X(x)$$

- Suppose X is a continuous r.v. with p.d.f. $f_X(x)$, then the variance of $g(X)$ is defined as

$$\text{Var}(g(X)) = \int_{-\infty}^{\infty} (g(x) - \mathbb{E}[g(X)])^2 f_X(x) dx$$

- Alternative: $\text{Var}(g(X)) = \mathbb{E}[g(X)^2] - \mathbb{E}[g(X)]^2$

- Properties of variance:

- $\text{Var}(a) = 0$ for any constant $a \in \mathbb{R}$
- $\text{Var}(aX) = a^2\text{Var}(X)$ for any constant $a \in \mathbb{R}$
- Linearity of variance:
 $\text{Var}(ag(X) + bf(Y)) = a^2\text{Var}(g(X)) + b^2\text{Var}(f(Y))$ for any constant $a, b \in \mathbb{R}$ and X and Y are independent

After-class Practice

- Calculate the expectations and variances of the following r.v.
 - The resulting number of tossing a die
 - The number of heads in an experiment that a fair coin is flipped 3 times
 - A uniform distributed r.v. X with p.d.f. $f_X(x) = 1$, where $0 \leq x \leq 1$

Common Discrete Distributions

	p.m.f.	Domain	$E[X]$	$\text{Var}(X)$
$\text{Bern}(\omega)$	$p_X(x) = \omega^x (1 - \omega)^{1-x}, \quad 0 \leq \omega \leq 1$	$\{0, 1\}$	ω	$\omega(1 - \omega)$
$\text{Bin}(N, \omega)$	$p_X(x) = \binom{N}{x} \omega^x (1 - \omega)^{N-x}$ $N > 0, 0 \leq \omega \leq 1$	$\{0, \dots, N\}$	$N\omega$	$N\omega(1 - \omega)$
$\text{Ge}(\omega)$	$p_X(x) = (1 - \omega)^{x-1} \omega, \quad 0 \leq \omega \leq 1$	$\mathbb{N} \setminus \{0\}$	$1/\omega$	$(1 - \omega)/\omega^2$
$\text{NB}(r, \omega)$	$p_X(x) = \binom{x+r-1}{x} (1 - \omega)^r \omega^x$ $r > 0, 0 \leq \omega \leq 1$	$\mathbb{N}$	$r\omega/(1 - \omega)$	$r\omega/(1 - \omega)^2$
$\text{NB}(\lambda, \phi)$	$p_X(x) = \frac{\Gamma(\phi+x)}{x! \Gamma(\phi)} \left(\frac{\lambda}{\phi+\lambda}\right)^x \left(\frac{\phi}{\phi+\lambda}\right)^\phi,$ $\lambda > 0, \phi > 0$	$\mathbb{N}$	λ	$\lambda + \lambda^2/\phi$
$\text{Poi}(\lambda)$	$p_X(x) = \exp(-\lambda) \lambda^x / x!, \quad \lambda > 0$	$\mathbb{N}$	λ	λ

Note that $\Gamma(x) = \begin{cases} (x-1)! & \text{if } x \in \mathbb{N} \\ \int_0^\infty y^{x-1} \exp(-y) dy & \text{if } x \in \mathbb{R}_+ \end{cases}$

Common Continuous Distributions

	p.d.f.	Domain	$E[X]$	$\text{Var}(X)$
$\text{Unif}(\alpha, \beta)$	$f_X(x) = \frac{1}{\beta - \alpha}, \quad \alpha < \beta$	$[\alpha, \beta]$	$\frac{\alpha + \beta}{2}$	$\frac{(\beta - \alpha)^2}{12}$
$\text{Be}(\alpha, \beta)$	$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1},$ $\alpha > 0, \beta > 0$	$[0, 1]$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$
$\text{Exp}(\lambda)$	$f_X(x) = \lambda \exp(-\lambda x), \quad \lambda > 0$	$\mathbb{R}_+$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\text{Ga}(\alpha, \beta)$	$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x),$ $\alpha > 0, \beta > 0$	$\mathbb{R}_+$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
$\text{IG}(\alpha, \beta)$	$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp(-\beta/x),$ $\alpha > 0, \beta > 0$	$\mathbb{R}_+$	$\frac{\beta}{\alpha - 1}$	$\frac{\beta^2}{(\alpha - 1)^2(\alpha - 2)}$
$\text{N}(\mu, \sigma^2)$	$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad \sigma > 0$	$\mathbb{R}$	μ	σ^2
$t(\nu, \mu, \sigma)$	$f_X(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\pi\nu\sigma^2}} \left(1 + \frac{1}{\nu} \frac{(x-\mu)^2}{\sigma^2}\right)^{-\frac{\nu+1}{2}},$ $\nu > 0, \sigma > 0$	$\mathbb{R}$	μ	$\sigma^2 \frac{\nu}{\nu-2}$

Note that $\Gamma(x) = \begin{cases} (x-1)! & \text{if } x \in \mathbb{N} \\ \int_0^\infty y^{x-1} \exp(-y) dy & \text{if } x \in \mathbb{R}_+ \end{cases}$

Binomial Random Variable

- Counts the number of successes in N Bernoulli trials, where $X \in \{0, 1, \dots, N\}$
- Two parameters:
 - The number of trials N , $N \in \mathbb{N}^+$
 - The success probability ω , $0 \leq \omega \leq 1$
- The p.m.f. is

$$p_X(x; N, \omega) = \binom{N}{x} \omega^x (1 - \omega)^{N-x},$$

where the term $\binom{N}{x} = \frac{N!}{(N-x)!x!}$ is called the binomial coefficient

- The expectation $E[X] = N\omega$ and the variance $\text{Var}(X) = N\omega(1 - \omega)$

Binomial Random Variable

- Examples:

- Flip $N = 10$ fair coins (i.e. $\omega = 0.5$), and let X be the number of heads: $X \sim \text{Bin}(10, 0.5)$
- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls: $X \sim \text{Bin}(100, 0.6)$
- Ask $N = 1,000$ persons in a public-opinion poll, where 80% of the population would respond favorably to the question were they asked, and let X be the observed number of people responding “Yes”: $X \sim \text{Bin}(1000, 0.8)$

Binomial Random Variable in R

- The arguments `size` and `prob` correspond to the parameters N and ω , respectively
- The function `dbinom(x, size, prob)` return the p.m.f.
- The function `pbinom(x, size, prob)` return the c.d.f.
- The function `qbinom(p, size, prob)` return the quantiles
- The function `rbinom(n, size, prob)` generates random samples

Binomial Random Variable in R

- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls:

$$X \sim \text{Bin}(100, 0.6)$$

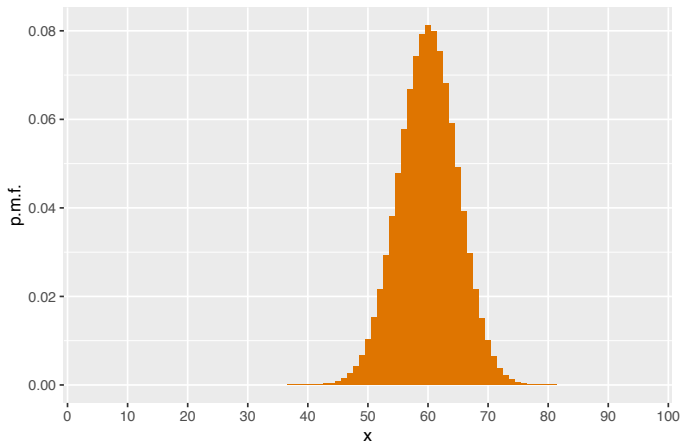
- How does the p.m.f. look?

```
x <- 0:100
p <- dbinom(x, size = 100, prob = 0.6)

data <- data.frame(x = x, p = p)
ggplot(data) + geom_bar(mapping = aes(x = as.
  factor(x), y = p), stat = "identity", fill =
  rgb(223/255, 117/255, 0)) +
  scale_x_discrete(breaks = seq(0, 100, by = 10))
  +
  labs(x = "x", y = "p.m.f.")
```

Binomial Random Variable in R

- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls:
 $X \sim \text{Bin}(100, 0.6)$
 - How does the p.m.f. look?



Binomial Random Variable in R

- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls:

$$X \sim \text{Bin}(100, 0.6)$$

- What are the quartiles?

```
qbinom(c(0.25, 0.5, 0.75), size = 100, prob = 0.6)
# The answers are 57, 60, and 63
```

Binomial Random Variable in R

- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls:

$$X \sim \text{Bin}(100, 0.6)$$

- Generate 1,000 r.v.'s and compare their histogram with the p.m.f.

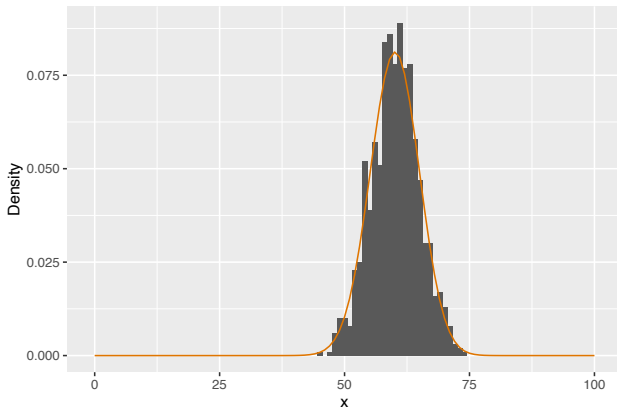
```
x <- 0:100
p <- dbinom(x, size = 100, prob = 0.6)
data <- data.frame(x = x, p = p)

x_sim <- rbinom(1000, size = 100, prob = 0.6)
data_2 <- data.frame(x = x_sim)

ggplot() + geom_histogram(data = data_2,
  mapping = aes(x = x_sim, y = ..density..),
  binwidth = 1) + geom_line(data = data,
  mapping = aes(x = x, y = p))
```

Binomial Random Variable in R

- Draw $N = 100$ balls from a box in which 60% balls are red and the rest are blue, and let X be the number of red balls:
 $X \sim \text{Bin}(100, 0.6)$
 - Generate 1,000 r.v.'s and compare their histogram with the p.m.f.



Normal Random Variable

- Represent real-valued measurements whose distributions are not known but bell-shaped, where $X \in \mathbb{R}$
- Two parameters:
 - The center μ , $\mu \in \mathbb{R}$
 - The spread σ^2 , $\sigma \in \mathbb{R}_+$
- The p.d.f. is

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- The expectation $E[X] = \mu$ and the variance $\text{Var}(X) = \sigma^2$

Normal Random Variable

- Examples:

- The mean of a male's height is 70.2 inches (i.e. $\mu = 70.2$) and the standard deviation is 2.89 inches (i.e. $\sigma = 2.89$) , let X be the height of a randomly selected male in the population: $X \sim N(70.2, 2.89^2)$
- The mean of the SAT is 1,059 (i.e. $\mu = 1,059$) and the standard deviation is 210 (i.e. $\sigma = 210$) , let X be the score of a randomly selected student who takes SAT: $X \sim N(1,059, 210^2)$

Normal Random Variable in R

- The arguments `mean` and `sd` correspond to the parameters μ and σ , respectively
- The function `dnorm(x, mean, sd)` return the p.d.f.
- The function `pnorm(x, mean, sd)` return the c.d.f.
- The function `qnorm(p, mean, sd)` return the quantiles
- The function `rnorm(n, mean, sd)` generates random samples

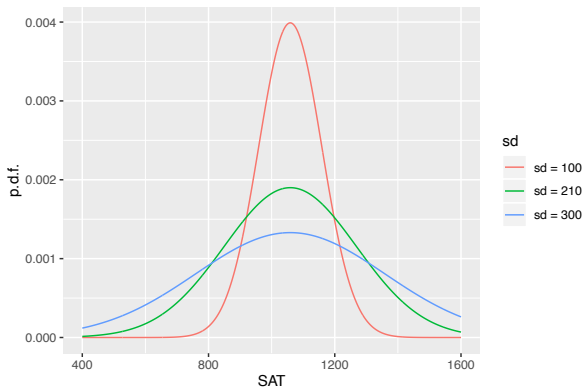
Normal Random Variable in R

- The mean of the SAT is 1,059 (i.e. $\mu = 1,059$) and the standard deviation is 210 inches (i.e. $\sigma = 210$), let X be the score of a randomly selected student who takes SAT: $X \sim N(1,059, 210^2)$
 - How does the p.d.f. look? Also try $\sigma = 150$ and $\sigma = 300$

```
x <- 400:1600
p_100 <- dnorm(x, mean = 1059, sd = 100)
p_210 <- dnorm(x, mean = 1059, sd = 210)
p_300 <- dnorm(x, mean = 1059, sd = 300)
data <- data.frame(x = rep(x, 3), p = c(p_100, p_210, p_300), sd = c(rep("sd = 100", 1201), rep("sd = 210", 1201), rep("sd = 300", 1201)))
ggplot(data) + geom_line(mapping = aes(x = x, y = p, color = sd, linetype = sd)) + labs(x = "SAT", y = "p.d.f.")
```

Normal Random Variable in R

- The mean of the SAT is 1,059 (i.e. $\mu = 1,059$) and the standard deviation is 210 (i.e. $\sigma = 210$), let X be the score of a randomly selected student who takes SAT: $X \sim N(1,059, 210^2)$
 - How does the p.d.f. look? Also try $\sigma = 150$ and $\sigma = 300$



After-class Practice

- The mean of the SAT is 1,059 (i.e. $\mu = 1,059$) and the standard deviation is 210 (i.e. $\sigma = 210$), let X be the score of a randomly selected student who takes SAT: $X \sim N(1,059, 210^2)$
 - The range of SAT score of students enrolled in UT Dallas in the year of 2016 is 1140 and 1380. What's the proportion of SAT taken students are in this range?
 - Generate 1,000 r.v.'s and compare their histogram with the p.d.f.

Common Distributions in R

	d function	p function	q function	r function
Bern(ω)	<code>dbinom(x, size = 1, prob = ω)</code>	<code>pbinom(x,)</code>	<code>qbinom(x,)</code>	<code>rbinom(n,)</code>
Bin(N, ω)	<code>dbinom(x, size = N, prob = ω)</code>	<code>pbinom(x,)</code>	<code>qbinom(p,)</code>	<code>rbinom(n,)</code>
Ge(ω)	<code>dgeom(x+1, prob = ω)</code>	<code>pgeom(x+1,)</code>	<code>1+qgeom(p,)</code>	<code>1+rgeom(n,)</code>
NB(r, ω)	<code>dnbinom(x, size = r, prob = ω)</code>	<code>pnbinom(x,)</code>	<code>qnbinom(p,)</code>	<code>rnbinom(n,)</code>
NB(λ, ϕ)	<code>dnbinom(x, mu = λ, size = ϕ)</code>	<code>pnbinom(x,)</code>	<code>qnbinom(p,)</code>	<code>rnbinom(n,)</code>
Poi(λ)	<code>dpois(x, mu = λ)</code>	<code>ppois(x,)</code>	<code>qpois(p,)</code>	<code>rpois(n,)</code>
Unif(α, β)	<code>dunif(x, min = α, max = β)</code>	<code>punif(x,)</code>	<code>qunif(p,)</code>	<code>runif(n,)</code>
Be(α, β)	<code>dbeta(x, shape1 = α, shape2 = β)</code>	<code>pbeta(x,)</code>	<code>qbeta(p,)</code>	<code>rbeta(n,)</code>
Exp(λ)	<code>dexp(x, rate = λ)</code>	<code>pexp(x,)</code>	<code>qexp(p,)</code>	<code>rexp(n,)</code>
Ga(α, β)	<code>dgamma(x, shape1 = α, rate = β)</code>	<code>pgamma(x,)</code>	<code>qgamma(p,)</code>	<code>rgamma(n,)</code>
IG(α, β)	<code>dgamma(1/x, shape1 = α, rate = β)</code>	<code>1-pgamma(1/x,)</code>	<code>1/qgamma(1-p,)</code>	<code>1/rgamma(n,)</code>
N(μ, σ^2)	<code>dnorm(x, mean = μ, sd = σ)</code>	<code>pnorm(x,)</code>	<code>qnorm(p,)</code>	<code>rnorm(n,)</code>
LN(μ, σ^2)	<code>dlnorm(x, meanlog = μ, sdlog = σ)</code>	<code>plnorm(x,)</code>	<code>qlnorm(p,)</code>	<code>rlnorm(n,)</code>
t(ν, μ, σ)	<code>dt((x-μ)/σ, df = ν)</code>	<code>pt(x-μ)/σ,)</code>	<code>qt(p,)*σ+μ</code>	<code>rt(n,)*σ+μ</code>

- Suppose a , b , and c are positive constants and X is a r.v. that satisfies
 - $f_X(x; a, b) \propto x^a(1 - x)^b$
 - $f_X(x; a, b) \propto x^a \exp(-bx)$
 - $f_X(x; a, b) \propto \exp\left(-\frac{1}{2}ax^2 + bx\right)$
 - $f_X(x; a, b, c) \propto \left(1 + \frac{(x-b)^2}{c}\right)^{-\frac{a+1}{2}}$
- Then use R to generate X , when a , b , and c are given
 - `rbeta(1, shape1 = a + 1, shape2 = b + 1)`
 - `rgamma(1, shape1 = a + 1, rate = b)`
 - `rnorm(1, mean = b/a, sd = sqrt(1/a))`
 - `rt(1, df = a) * c / a + b`

Joint Cumulative Distribution Function

- Two r.v.'s X and Y (usually not independent)
- The joint c.d.f. is a function $F_{XY} : \mathbb{R}^2 \rightarrow [0, 1]$ such that

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

- The marginal c.d.f. $F_X, F_Y : \mathbb{R} \rightarrow [0, 1]$ can be derived from the joint c.d.f. F_{XY}
 - $F_X(x) = \lim_{y \rightarrow \infty} F_{XY}(x, y)$
 - $F_Y(y) = \lim_{x \rightarrow \infty} F_{XY}(x, y)$
- $F_{XY}(x, y)$ should satisfy:
 - $0 \leq F_{XY}(x, y) \leq 1$
 - $\lim_{x, y \rightarrow -\infty} F_{XY}(x, y) = 0$
 - $\lim_{x, y \rightarrow \infty} F_{XY}(x, y) = 1$

Joint Probability Mass Function

- Both r.v.'s X and Y takes on a finite set of possible values
- The joint p.m.f. is a function $p_{XY} : \mathbb{N}^2 \rightarrow [0, 1]$ such that

$$p_{XY}(x, y) = P(X = x, Y = y)$$

- The marginal p.m.f. $p_X, p_Y : \mathbb{N} \rightarrow [0, 1]$ can be derived from the joint p.m.f. p_{XY}
 - $p_X(x) = \sum_{y \in \text{Val}(Y)} p_{XY}(x, y)$
 - $p_Y(y) = \sum_{x \in \text{Val}(X)} p_{XY}(x, y)$
 - Form the marginal distribution w.r.t. one variable by summing out the other variable
- $p_{XY}(x, y)$ should satisfy:
 - $0 \leq p_{XY}(x, y) \leq 1$
 - $\sum_{x \in \text{Val}(X)} \sum_{y \in \text{Val}(Y)} p_{XY}(x, y) = 1$

Joint Probability Density Function

- Both r.v.'s X and Y takes on continuous values
- The joint p.d.f. is a function $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ such that

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y},$$

if $F_{XY}(x, y)$ is differentiable in both x and y

- The marginal p.d.f. $f_X, f_Y : \mathbb{R} \rightarrow \mathbb{R}_+$ can be derived from the joint p.d.f. f_{XY}
 - $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$
 - $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$
- $f_{XY}(x, y)$ should satisfy:
 - $f_{XY}(x, y) \geq 0$
 - $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) = 1$

Conditional Distribution

- The distribution of one variable Y , given that the other variable takes on a certain value X
- The conditional p.m.f. $p_{Y|X} : \mathbb{N} \rightarrow [0, 1]$ such that

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)}$$

- The conditional p.d.f. $f_{Y|X} : \mathbb{R} \rightarrow \mathbb{R}_+$ such that

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

- The conditional c.d.f. $F_{Y|X} : \mathbb{R} \rightarrow [0, 1]$ such that

$$F_{Y|X}(y|x) = \int_{-\infty}^y \frac{f_{XY}(x, z)}{f_X(x)} dz$$

Independence

- The value of one variable Y has no effect on the value of the other variable X , and *vice versa*.
- Two r.v.'s X and Y are independent if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

- For discrete r.v.'s verify

$$p_{XY}(x, y) = p_X(x)p_Y(y) \text{ or } p_{Y|X}(y|x) = p_Y(y)$$

- For continuous r.v.'s verify

$$f_{XY}(x, y) = f_X(x)f_Y(y) \text{ or } f_{Y|X}(y|x) = f_Y(y)$$

Bayes' Theorem

- For discrete r.v.'s

$$p_{Y|X}(y|x) = \frac{p_{XY}(x,y)}{p_X(x)} = \frac{p_{X|Y}(x|y)p_Y(y)}{\sum_{z \in \text{Val}(Y)} p_{X|Y}(x|z)p_Y(z)}$$

- For continuous r.v.'s

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{X|Y}(x|y)f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x|z)f_Y(z)dz}$$

Expectation

- Suppose $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is an arbitrary function
- Suppose X and Y are two discrete r.v.'s with joint p.m.f. p_{XY} , then the expectation of $g(X, Y)$ is defined as

$$E[g(X, Y)] = \sum_{x \in \text{Val}(X)} \sum_{y \in \text{Val}(Y)} g(x, y) p_{XY}(x, y)$$

- Suppose X and Y are two continuous r.v.'s with p.d.f. f_{XY} , then the expectation of $g(X, Y)$ is defined as

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy$$

- Suppose X and Y are two r.v's, then the covariance of X and Y is defined as

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - XE[Y] - YE[X] - E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

- Properties of covariance:
 - $\text{Cov}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$ for any constant $a, b \in \mathbb{R}$
 - X and Y are independent $\Rightarrow \text{Cov}(X, Y) = 0$
 - $\text{Cov}(X, Y) = 0 \Rightarrow X$ and Y are uncorrelated

- Let $X \sim \text{Unif}(-1, 1)$ and $Y = \begin{cases} X & \text{if } X > 0 \\ -X & \text{if } X \leq 0 \end{cases}$
 - X and Y are dependent!
- $X \sim \text{Unif}(-1, 1)$, $f_X(x) = 1/2$, $-1 \leq x \leq 1$, and $E[X] = 0$
- $Y \sim \text{Unif}(0, 1)$, $f_Y(y) = 1$, $0 \leq y \leq 1$, and $E[Y] = 1/2$
- $E[XY|X > 0] = E[X^2|X > 0] = \int_0^1 x^2 f_X(x) dx = 1/6$,
 $E[XY|X \leq 0] = E[-X^2|X \leq 0] = \int_{-1}^0 -x^2 f_X(x) dx = -1/6$, and
 $E[XY] = E[XY|X > 0]P(X > 0) + E[XY|X \leq 0]P(X \leq 0) = 0$
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$
 - X and Y are uncorrelated!

Multiple Random Variables

- More than two quantities of interest (not independent)
- The joint c.d.f. is a function $F_{X_1, \dots, X_p} : \mathbb{R}^p \rightarrow [0, 1]$ such that

$$F_{X_1, \dots, X_p}(x_1, \dots, x_p) = P(X_1 \leq x_1, \dots, X_p \leq x_p)$$

- The joint p.m.f. is a function $p_{X_1, \dots, X_p} : \mathbb{N}^p \rightarrow \mathbb{R}_+$ such that

$$p_{X_1, \dots, X_p}(x_1, \dots, x_p) = P(X_1 = x_1, \dots, X_p = x_p)$$

- The joint p.d.f. is a function $f_{X_1, \dots, X_p} : \mathbb{R}^p \rightarrow \mathbb{R}_+$ such that

$$f_{X_1, \dots, X_p}(x_1, \dots, x_p) = \frac{\partial^p F_{X_1, \dots, X_p}(x_1, \dots, x_p)}{\partial x_1 \dots \partial x_p},$$

if $F_{X_1, \dots, X_p}(x_1, \dots, x_p)$ is differentiable in every dimension

Multiple Random Variables

- The marginal p.d.f. is a function $f_{X_1} : \mathbb{R} \rightarrow \mathbb{R}_+$ such that

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_1, \dots, X_p}(x_1, \dots, x_p) dx_2 \dots dx_p$$

- The conditional p.d.f. is a function $f_{X_1|X_2, \dots, X_p} : \mathbb{R} \rightarrow \mathbb{R}_+$ such that

$$f_{X_1|X_2, \dots, X_p}(x_1|x_2, \dots, x_p) = \frac{f_{X_1, \dots, X_p}(x_1, \dots, x_p)}{f_{X_2, \dots, X_p}(x_2, \dots, x_p)}$$

- Chain rule:

$$\begin{aligned} f(x_1, \dots, x_p) &= f(x_1|x_2, \dots, x_p)f(x_2, \dots, x_p) \\ &= f(x_1|x_2, \dots, x_p)f(x_2|x_3, \dots, x_p)f(x_3, \dots, x_p) \\ &\vdots \\ &= f(x_1|x_2, \dots, x_p)f(x_2|x_3, \dots, x_p) \dots f(x_{p-1}|x_p)f(x_p) \end{aligned}$$

Multinomial Distribution

- A generalization of binomial distribution

- Binomial distribution:

$$p_X(x|N, \omega) = \binom{N}{x} \omega^x (1 - \omega)^{N-x} = \frac{N!}{x_1! x_2!} \omega_1^{x_1} \omega_2^{x_2},$$

where $x_1 := x$, $x_2 := N - x$, $\omega_1 := \omega$, and $\omega_2 := 1 - \omega$

- Multinomial distribution:

$$p_{\mathbf{X}}(\mathbf{x}|N, \omega_1, \dots, \omega_p) = \frac{N!}{x_1! \dots x_p!} \prod_{j=1}^p \omega_j^{x_j},$$

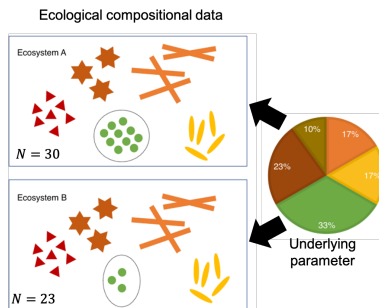
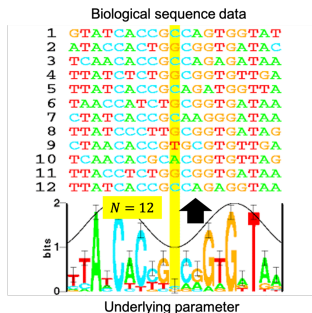
where $x_j \in \{0, \dots, N\}$, $\sum_{j=1}^p x_j = N$ and
 $N \in \mathbb{N} \setminus \{0\}$, $\omega_j \geq 0$, $\sum_{j=1}^p \omega_j = 1$

Multinomial Distribution

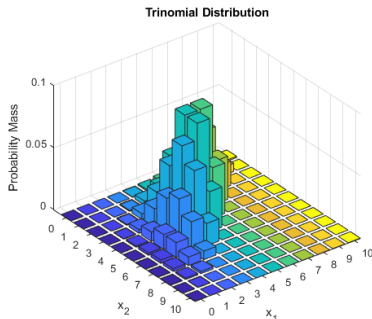
- Sample x

- Observed data: x , where $x = (x_1, \dots, x_p)$ subjecting to $\sum_{j=1}^p x_j = N$
- A single observation from a population
- Sample size $n = 1$

- Examples



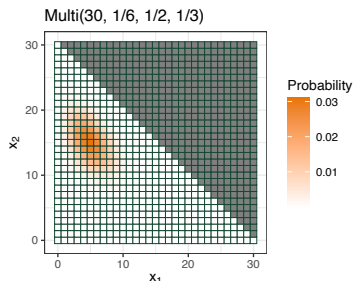
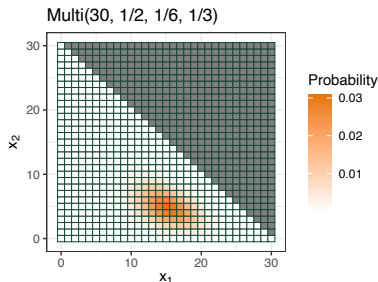
Multinomial Distribution



- Marginal distribution:
 - 1-dimension case: $X_j \sim \text{Bin}(N, \omega_j), j = 1, \dots, p$
- Conditional distribution: $X_1, \dots, X_{p-1} | X_p = x_p \sim \text{Multi} \left(N - x_p, \omega_1 / \sum_{j'=1}^{p-1} \omega'_{j'}, \dots, \omega_{p-1} / \sum_{j'=1}^{p-1} \omega'_{j'} \right)$

Multinomial Distribution in R

- `dmultinom(x, size = NULL, prob =  $\omega$ )`
- `rmultinom(n, size =  $N$ , prob =  $\omega$ )`



Multivariate Normal Distribution

- A generalization of normal distribution:
 - Normal distribution:

$$f_X(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right),$$

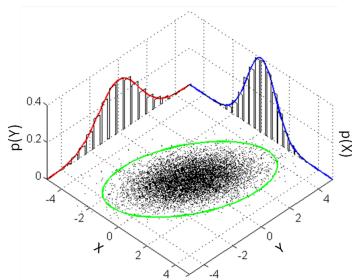
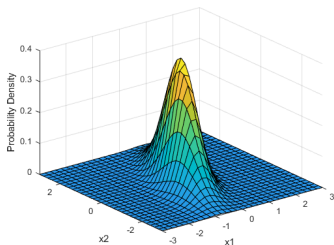
where $x \in \mathbb{R}$ and $\mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+$

- Multivariate normal distribution:

$$f_{\mathbf{X}}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^p |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right),$$

where $\mathbf{x} \in \mathbb{R}^p$ and $\boldsymbol{\mu} \in \mathbb{R}^p, \mathbf{y}^\top \boldsymbol{\Sigma} \mathbf{y} > 0, \forall$ non-zero $\mathbf{y}$

Multivariate Normal Distribution



- Marginal distribution:

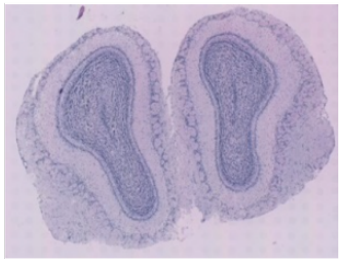
- 1-dimension case: $X_j \sim N(\mu_j, \sigma_j^2), j = 1, \dots, p$
- p' -dimension case ($1 < p' < p$): Drop the variables that one wants to marginalize out from both μ and Σ

- Conditional distribution (2-dimension case):

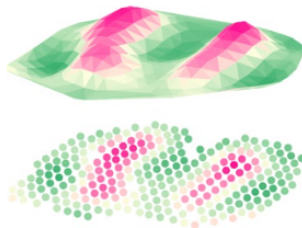
$X_1|X_2 = x_2 \sim N\left(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), (1 - \rho^2)\sigma_1^2\right)$, where ρ is the correlation coefficient between X_1 and X_2

Multivariate Normal Distribution

- Gaussian process (two-dimensional space)



Spatial expression pattern



108 genes



42 genes



59 genes



65 genes



41 genes



19 genes



Multivariate Normal Distribution

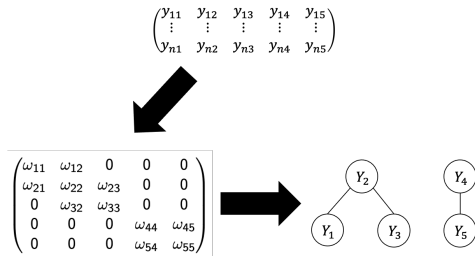
- Graphics:

$$G = (V, E)$$

- Gaussian graphical model:

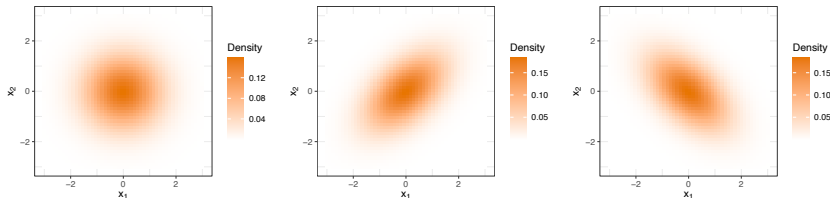
$$Y_i \perp\!\!\!\perp Y_j \quad \Leftrightarrow \quad \omega_{ij} = \omega_{ji} = 0$$

- Example



Multivariate Normal Distribution in R

- mvnfast package
- `dmvn(x, mu =  $\mu$ , sigma =  $\Sigma$ )`
- `rmvn(n, mu =  $\mu$ , sigma =  $\Sigma$ , ncores = 4)`



- `dmvt(x, df, mu =  $\mu$ , sigma =  $\Sigma$ )`
- `rmvt(n, df, mu =  $\mu$ , sigma =  $\Sigma$ , ncores = 4)`

In-class Practice: Basketball Free Throws

A basketball player makes free throws with probability

$$p = 0.75.$$

Suppose the player attempts 8 free throws.

Let

X = number of made free throws.

❶ What is the distribution of X ?

❷ What is

$$P(X = 6)?$$

❸ What is

$$P(X \geq 6)?$$

In-class Practice: Basketball Free Throws

Since each free throw has two possible outcomes,

$$X_i \sim \text{Bernoulli}(0.75).$$

With 8 independent attempts,

$$X \sim \text{Binomial}(8, 0.75).$$

Therefore,

$$P(X = 6) = \binom{8}{6} (0.75)^6 (0.25)^2.$$

And

$$P(X \geq 6) = \sum_{x=6}^8 \binom{8}{x} (0.75)^x (0.25)^{8-x}.$$

In-class Practice: Basketball in R

Calculate $P(X = 6)$:

```
dbinom(6, size = 8, prob = 0.75)
```

Calculate $P(X \geq 6)$:

```
1 - pbinom(5, size = 8, prob = 0.75)
```

In-class Practice: Baseball

A baseball player gets a hit with probability

$$p = 0.30$$

in each at-bat.

Suppose the player has 10 at-bats.

Let

X = number of hits.

Find:

- 1 The distribution of X .
- 2 $P(X = 3)$.
- 3 $P(X \geq 3)$.
- 4 $E(X)$.
- 5 $\text{Var}(X)$.

In-class Practice: Baseball

We have

$$X \sim \text{Binomial}(10, 0.30).$$

Therefore,

$$P(X = 3) = \binom{10}{3} (0.30)^3 (0.70)^7.$$

The expected number of hits is

$$E(X) = 10(0.30) = 3.$$

The variance is

$$\text{Var}(X) = 10(0.30)(0.70).$$

The expected number of hits is 3, but the player will not necessarily get exactly 3 hits.

In-class Practice: Tennis

A tennis player wins a point on

70%

of first serves.

Suppose the player hits 5 first serves.

Let

X = number of points won.

Answer:

- 1 What is the distribution of X ?
- 2 What is $P(X = 4)$?
- 3 What is $P(X \geq 4)$?

In-class Practice: Geometric Distribution

A basketball player makes a three-point shot with probability

$$p = 0.40.$$

Let

X = number of attempts until the first made three-pointer.

Find

$$P(X = 1), \quad P(X = 2), \quad P(X = 3).$$

What is the probability that the player misses the first three attempts?

$$P(X > 3) = ?$$

In-class Practice: Poisson Goals

Suppose a soccer team scores an average of

$$\lambda = 1.8$$

goals per game.

Assume

$$X \sim \text{Poisson}(1.8).$$

Find:

- ① $P(X = 0)$
- ② $P(X = 1)$
- ③ $P(X = 2)$
- ④ $P(X \geq 3)$

In-class Practice: Poisson in R

Calculate the probability of exactly two goals:

```
dpois(2, lambda = 1.8)
```

Calculate the probability of at least three goals:

```
1 - ppois(2, lambda = 1.8)
```

Generate one simulated game:

```
rpois(1, lambda = 1.8)
```

In-class Practice: Normal Distribution

Suppose pitch velocity is approximately

$$X \sim N(95, 2^2).$$

Find:

- ① $P(X > 98)$
- ② $P(X < 92)$
- ③ $P(93 < X < 97)$

What does the standard deviation of 2 mph tell us about pitch velocity?

In-class Practice: Normal Distribution in R

Calculate $P(X > 98)$:

```
1 - pnorm(98, mean = 95, sd = 2)
```

Calculate $P(93 < X < 97)$:

```
pnorm(97, mean = 95, sd = 2) -  
pnorm(93, mean = 95, sd = 2)
```

In-class Practice: Expected Points

Suppose a basketball possession ends with

Outcome	Probability
0 points	0.50
2 points	0.35
3 points	0.15

Let X be points scored on the possession.

Find

$$E(X).$$

$$E(X) = 0(0.50) + 2(0.35) + 3(0.15).$$

Therefore,

$$E(X) = 1.15.$$

In-class Practice: Player Consistency

Two basketball players both average

20

points per game.

Player A:

$$SD_A = 3.$$

Player B:

$$SD_B = 10.$$

Questions:

- 1 Who has greater variability?
- 2 Who is more consistent?
- 3 Can two players have the same mean but very different distributions?

In-class Practice: Home Court Advantage

Let

$$W = \{\text{team wins}\}$$

and

$$H = \{\text{team plays at home}\}.$$

Suppose

$$P(W) = 0.60, \quad P(H) = 0.50,$$

and

$$P(W \cap H) = 0.35.$$

Find

$$P(W \mid H).$$

In-class Practice: Independence in Sports

Using the previous example,

$$P(W) = 0.60, \quad P(H) = 0.50,$$

and

$$P(W \cap H) = 0.35.$$

Are W and H independent?

If they were independent, then

$$P(W \cap H) = P(W)P(H) = 0.60(0.50) = 0.30.$$

But

$$0.35 \neq 0.30.$$

Therefore,

In-class Practice: Bayes' Theorem

Suppose a team is either strong or weak.

Let

$$S = \{\text{strong team}\}.$$

Suppose

$$P(S) = 0.30.$$

A strong team has probability

$$P(D \mid S) = 0.80$$

of winning three consecutive games.

A weak team has probability

$$P(D \mid S^c) = 0.20.$$

After observing three wins, calculate

In-class Practice: Bayes' Theorem

Using Bayes' theorem,

$$P(S | D) = \frac{P(D | S)P(S)}{P(D)}.$$

The denominator is

$$P(D) = P(D | S)P(S) + P(D | S^c)P(S^c).$$

Therefore,

$$P(S | D) = \frac{0.80(0.30)}{0.80(0.30) + 0.20(0.70)}.$$

Thus,

$$P(S | D) \approx 0.632.$$

In-class Practice: Interpret Bayes' Theorem

Before observing the games,

$$P(S) = 0.30.$$

After observing three consecutive wins,

$$P(S \mid D) \approx 0.632.$$

What happened?

$$\boxed{0.30 \longrightarrow 0.632}$$

The observed results increased our belief that the team is strong.

This is the basic idea of Bayesian updating.

In-class Practice: Player Ability

Suppose a basketball player has an unknown three-point shooting probability p .

The player makes

60

out of

100

attempts.

The observed shooting percentage is

$$\hat{p} = \frac{60}{100} = 0.60.$$

But ask:

$$P(p > 0.50 \mid Y = 60) = ?$$

In-class Practice: Prior Information

Suppose

$$p \sim \text{Beta}(10, 10).$$

Then

$$E(p) = \frac{10}{20} = 0.50.$$

The sampling model is

$$Y \mid p \sim \text{Binomial}(100, p).$$

After observing

$$Y = 60,$$

what is the posterior distribution?

$$p \mid Y = 60 \sim \text{Beta}(70, 50).$$

In-class Practice: More Data

Player A makes

$$6/10 = 0.60.$$

Player B makes

$$600/1000 = 0.60.$$

Suppose they have the same prior distribution.

Questions:

- 1 Do they have the same posterior mean?
- 2 Do they have the same posterior uncertainty?
- 3 Which player's ability is estimated more precisely?

Same estimate does not mean same amount of information.

In-class Practice: Sports Data Types

Match the sports question with a suitable probability model.

Question	Possible Model
Made or missed shot	Bernoulli
Hits in 100 at-bats	Binomial
Goals in a soccer game	Poisson
Pitch velocity	Normal
Unknown shooting probability	Beta
Several possession outcomes	Multinomial

In-class Practice: Probability or Inference?

Classify each question.

- ① A player makes a shot with probability 0.70. What is the probability of making 8 of 10 shots?
- ② A player made 8 of 10 shots. What can we learn about the player's true shooting ability?
- ③ A team wins 60% of games. What is the probability of winning at least 50 of 82 games?
- ④ A team wins 50 of 82 games. How strong is the team?

The first and third are primarily **probability** questions.

The second and fourth move toward **statistical inference**.

In-class Practice: Simulation and Randomness

Suppose a team has true win probability

$$p = 0.60.$$

Simulate 82 games.

How many wins would you expect?

$$E(W) = 82(0.60) = 49.2.$$

Does this mean every simulated season will have exactly 49 or 50 wins?

No. Randomness creates season-to-season variation.

In-class Practice: Simulate a Season

Simulate 10,000 seasons:

```
wins <- rbinom(10000, size = 82, prob = 0.60)
```

Then calculate:

```
mean(wins)
sd(wins)
quantile(wins, c(0.025, 0.5, 0.975))
```

Question:

What does the simulated distribution tell us about the variability of an 82-game season?

Key Takeaway from the Sports Examples

Across these examples, the workflow is the same:

Define the random variable



Choose a probability model



Calculate probabilities



Use data to learn about unknown quantities

After-class Reading

- *A First Course in Bayesian Statistical Methods* by Peter Hoff
 - Chapter 2 Belief, probability and exchangeability
 - Section 2.4 Random variables
 - Section 2.5 Joint distributions
 - Section 2.6 Independent random variables